Graph Transformer Geometric Learning of Brain Networks Using Multimodal MR Images for Brain Age Estimation

Hongjie Cai, Yue Gao, *Member, IEEE*, and Manhua Liu, *Member, IEEE*

***Abstract*— Brain age is considered as an important biomarker for detecting aging-related diseases such as Alzheimer's Disease (AD). Magnetic resonance imaging (MRI) have been widely investigated with deep neural networks for brain age estimation. However, most existing methods cannot make full use of multimodal MRIs due to the difference in data structure. In this paper, we propose a graph transformer geometric learning framework to model the multimodal brain network constructed by structural MRI (sMRI) and diffusion tensor imaging (DTI) for brain age estimation. First, we build a two-stream convolutional autoencoder to learn the latent representations for each imaging modality. The brain template with prior knowledge is utilized to calculate the features from the regions of interest (ROIs). Then, a multi-level construction of the brain network is proposed to establish the hybrid ROI connections in space, feature and modality. Next, a graph transformer network is proposed to model the cross-modal interaction and fusion by geometric learning for brain age estimation. Finally, the difference between the estimated age and the chronological age is used as an important biomarker for AD diagnosis. Our method is evaluated with the sMRI and DTI data from UK Biobank and Alzheimer's Disease Neuroimaging Initiative database. Experimental results demonstrate that our method has achieved promising performances for brain age estimation and AD diagnosis.**

***Index Terms*— Brain age estimation, multimodal MRI, graph transformer, brain network, Alzheimer's disease.**

I. Introduction

AGING is a complex process that goes with the gradual biological and physiological changes in an organism as the individual grows older. It also takes place in human brain with the passage of time, termed as brain aging. Brain age was defined as the biological age of an individual brain [1]. To capture the brain changes with aging, neuroimaging techniques such as magnetic resonance imaging (MRI) can provide a powerful tool to investigate the effects of age-related alterations in brain structure and function, including morphometric atrophy and connectivity reduction [1], [2]. Multimodal MRI including structural MRI (sMRI), function MRI (fMRI) and diffusion tensor imaging (DTI) can capture variant structural and functional information of living human brain under non-invasive or minimally invasive conditions and thus has been widely used in the studies of brain aging process and the clinical diagnosis of brain diseases [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18].

Across the late adult lifespan, although normal aging may cause the changes of brain, the risk of developing neurodegenerative disorders such as Alzheimer's disease (AD) rises with increasing age, partly as a result of pathological processes associated with accelerated brain aging [1]. To accurately predict the disease risk, it is important to measure the individual differences between the normal and disease-specific brain aging. One promising approach is to develop machine learning or deep learning models on the neuroimaging data to estimate the brain age [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. The difference between the estimated brain age and chronological age is defined as estimated age difference (EAD), which indicates the degree of deviation from the normal brain aging trajectory. A positive EAD is assumed to be an intuitive manifestation of accelerated brain aging, while a negative EAD means that brain aging is decelerated or healthy. EAD has been proved to be an important biomarker for the age-related brain diseases such as AD [1], [5], [6], [12]. AD is one of the most prevalent neurodegenerative diseases and mild cognitive impairment (MCI) is the transitional state from the healthy state, i.e., normal control (NC) to AD. Currently, there is no effective cure for AD, and early diagnosis is important for appropriate treatment of patients. Therefore, accurate brain age estimation and abnormal aging analysis are of great help for the early diagnosis of AD [5], [12].

To quantify the individual brain development, traditional machine learning methods have been employed for brain age estimation with the MR images of healthy subjects [3], [4], [5], [6], [7]. Generally, there are three fundamental steps in these

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methods: data preprocessing, feature extraction and age regression. For different imaging modalities, different pipelines are used for preprocessing of raw imaging data [6], [7]. After data preprocessing, variant measurements are computed to characterize the patterns related to brain aging. As for 3D sMRI data, the morphological features such as tissue intensities, volumes of ROIs (regions of interest), cortical thickness are extracted for brain age estimation [3], [6]. As for DTI data, fractional anisotropy (FA) map is obtained after data preprocessing, and the fiber tract-tracing algorithm is used to generate a connectivity matrix of brain ROIs as the features [4], [7]. Based on the features from MR images, brain age is estimated by multivariate regression models such as relevance vector regression [5], support vector regression (SVR) with random forest (RF) [6] and backpropagation neural network [7], etc.

Recently, deep learning has been successfully investigated on MR image analysis for brain age estimation because of its powerful capability in jointly learning image features and age regression [8], [9], [10], [11], [12], [13], [14], [15], [16]. Most of these methods are based on the architecture of convolutional neural network (CNN) to build 2D CNN on slices of sMRI scans [10] and 3D CNN on whole brain MRI data [8], [9], [13]. To be more compatible with small dataset, a simple 3D fully convolutional network (SFCN) with fewer parameters was proposed for brain age estimation using sMRI data [11]. A 3D two-stage-age-network (TSAN) was proposed with the first-stage network estimating a rough brain age, followed by a second-stage network for more accurate estimation [12]. A fusion-with-attention deep network (FiA-Net) was proposed to fuse the intensity and RAVENS channels of sMRI data for brain age estimation [14]. A global-local transformer was proposed to build the 2D CNNs on whole sMRI data and local patches to learn the global-context and local fine-grained features, respectively, which were combined by self-attention mechanism for brain age estimation [15]. In addition to the CNN architecture, graph neural network (GNN) has been explored to build the graphs of brain ROIs and model the brain network on MR images [16], [17]. A graph network was created with brain ROIs as nodes and ROI volume dependencies as connections for brain age estimation with sMRI data [16]. A graph-based approach was proposed to construct brain networks with the nodes representing brain regions and the edges for the connections of regions, which were used to explore the structural connectivity of brain aging on DTI data [17].

Multimodal MRI data can provide complementary information about brain structure and function and thus have been investigated to improve brain age estimation [4], [6], [18], [19], [20]. Since sMRI and DTI are two powerful imaging data for characterization of brain structure, the cortical thickness and structural connectivity were extracted and concatenated for multivariate regression of brain ages [4]. A stacked-multimodal approach was proposed to combine cortical anatomy from sMRI and whole-brain functional connectivity from Resting-state fMRI for improving brain age estimation [6]. Multimodal MRI data including T1-weighted, T2-relaxometry and DTI were investigated by a voxel-based multiple regression analysis for brain age estimation [18].

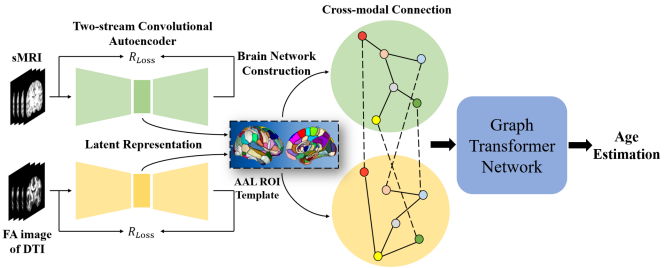

Fig. 1. Overview of our graph transformer geometric learning framework, consisting of a two-stream convolutional autoencoder for learning modality specific features, multi-level construction of brain networks, and a graph transformer network to learn representation for brain age estimation.

A multimodal brain age estimation model was proposed to combine the T1-weighted MRI, T2-FLAIR, T2*, diffusion-MRI, task fMRI, and resting-state fMRI with a LASSO regression [19]. The hand-crafted features are extracted using complex data preprocessing with prior knowledge, which are not jointly optimized with brain age regression. Recently, the SFCN with linear regression (LR) was employed to combine sMRI and time-of-flight magnetic resonance angiography (TOF MRA) for brain age estimation [20]. Although the CNN architecture has powerful ability to learn the features of MR images, the multimodal features are concatenated for final decision, ignoring the interactions between sMRI and DTI.

The sMRI can capture the variations of brain anatomy and morphometric atrophy while DTI captures information about the microstructural integrity of white matter [18]. They can provide essential and complementary biomarkers for characterization of brain structure for brain age estimation. However, it is still challenging to explore the interactions between different ROIs and different modalities for brain age estimation. To this end, we propose a graph transformer geometric learning framework to construct and analyze the multimodal brain networks of sMRI and DTI data for brain age estimation, as shown in Fig. 1. First, sMRI and DTI data are processed and fed into a two-stream convolutional autoencoder (CAE) for learning their latent representations. Second, the multi-level brain networks are constructed by ROI feature aggregation and diversified node connections in terms of different attributes (i.e., space, feature, and modality). Finally, a graph transformer network is proposed to model the interactions of intra- and inter-modalities and generate a graph representations for brain age estimation. Furthermore, the EAD is explored as an important biomarker for AD diagnosis. Our proposed method is trained and evaluated with sMRI and DTI data from a large cohort of subjects on two independent datasets.

TABLE I summaries the state-of-the-art models in recent studies for brain age estimation using MRI data with their mean absolute error (MAE) results reported in the literature. The MAE is around 2.1–5.6 years. Our method achieves MAE of 2.7 years, which is competitive and promising compared to other methods. Most methods are based on CNN architecture and sMRI data. Only a few studies have explored the fusion of multimodal MRI data for more accurate brain age estimation.

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TABLE I
A SUMMARY OF RECENT STUDIES FOR BRAIN AGE ESTIMATION USING MRI DATA OF HEALTHY SUBJECTS

Study	# Sample	Age Range	Modality	Method	MAE (years)
Liem et al., 2017 [6]	2354	19-82	sMRI + fMRI	SVR + RF	4.29
Cole, 2020 [19]	17426	45-80	sMRI, DTI etc.	LASSO regression	3.55
Feng et al., 2020 [8]	10158	18-97	sMRI	3D-CNN	4.06
Jiang et al., 2020 [9]	1454	18-90	sMRI	3D-CNN	5.55
Bashyam et al., 2020 [10]	11729	3-95	sMRI	2D-CNN	3.70
Peng et al., 2021 [11]	14503	42-82	sMRI	3D-SFCN	2.14
Cheng et al., 2021 [12]	6586	17-98	sMRI	3D-TSAN + Ranking Loss	2.43
Dinsdale et al., 2021 [13]	19687	44-80	sMRI	3D-CNN	2.97
He et al., 2021 [14]	16705	0-97	sMRI	3D-FiA-Net	3.00
He et al., 2021 [15]	8379	0-97	sMRI	2D-CNN + Transformer	2.70
Rajashekar et al., 2022 [20]	2074	21-81	sMRI + TOF MRA	3D-SFCN + LR	3.85
Proposed Method	**16458**	**46-81**	**sMRI + DTI**	**3D-CAE + Graph Transformer**	**2.71**

TABLE II
DEMOGRAPHIC INFORMATION OF THE STUDIED SUBJECTS FROM ADNI AND UKB

Dataset	Number	Age Range	Age Statistics	Male/Female
ADNI	151	55-90	73.5 ± 7.2	87 / 64
UKB	16458	46-81	63.9 ± 7.5	8572 / 7886

TABLE III
MORE DETAILED DEMOGRAPHIC AND CLINICAL INFORMATION OF THE STUDIED SUBJECTS FROM ADNI

Group	Number	Age Range	Age Statistics	MMSE	CDR
NC	43	62-89	73.0 ± 5.9	29.0 ± 1.1	0.0 ± 0.0
MCI	59	55-85	72.6 ± 6.6	26.1 ± 2.1	0.5 ± 0.1
AD	49	56-90	75.2 ± 8.5	24.0 ± 3.6	0.9 ± 0.3

Different from these studies, the main contributions of this paper are summarized as below:

(1) We propose a graph transformer geometric learning framework to build and combine the multimodal brain networks of sMRI and DTI, which can make full use of the local and global multimodal features for brain age estimation.

(2) We propose a multi-level construction of brain graph networks with diversified connections based on spatial relation, feature correlation and cross-modal association.

(3) A graph transformer network is proposed to model the cross-modal interaction and aggregate node features by geometric learning for more accurate brain age estimation.

The rest of this paper is organized as follows. In Section II and III, we introduce the studied materials and the proposed method in detail, respectively. In Section IV, we present experimental settings and results. Finally, we give a discussion and conclude this paper in Section V and VI, respectively.

II. MATERIALS

A. Datasets

In this study, we use two public datasets from UK Biobank (UKB) [21] and Alzheimer's Disease Neuroimaging Initiative (ADNI) [22]. UKB is a large-scale prospective cohort study of over 500,000 individuals from the United Kingdom. It collected the genetic data from each participant and some of the original participants were invited back for multimodal imaging of their brains, hearts and bodies. ADNI study collected a variety of multimodal brain images and clinical assessments, including sMRI, DTI and other biomarkers, from thousands of participants across North America for dementia research.

The UKB and ADNI datasets were downloaded from the websites https://biobank.ctsu.ox.ac.uk/ and http://adni.loni.usc.edu/, respectively. TABLE II summarizes the demographic information of the UKB and ADNI subjects in our study. This study collected the sMRI and DTI data of the 16458 healthy subjects from the UKB dataset. We did not use the subjects diagnosed with brain disorders such as AD in UKB because few of them have the multimodal MRI data. In contrast, ADNI is a commonly used dataset for AD study, which has explicit dementia rating labels. Thus, we collected the sMRI and DTI data of 151 subjects from ADNI dataset, which were diagnosed as three categories (i.e., AD, MCI and NC) based on the standard clinical criteria including mini-mental state examination (MMSE) score and clinical dementia rating (CDR). More detailed demographic and clinic information on the ADNI subjects in our study is shown in TABLE III.

Sufficient training data can make the deep learning model have a better generalization ability and avoid the overfitting problem. The number of healthy subjects with sMRI and DTI data from the ADNI dataset is much smaller than that of UKB. Thus, we train the proposed deep learning model with the large dataset of UKB and then transfer the trained model to ADNI dataset for brain age estimation and AD diagnosis.

B. Preprocessing

In this study, we download the sMRI and DTI data from the UKB dataset with the preprocessing and quality control pipeline as described in [23]. These data are fully preprocessed and available to all researchers granted access to UKB. This pipeline includes the distortion correction, cutting down the field of view, brain extraction and registration by using the FMRIB Software Library (FSL, https://fsl.fmrib.ox.ac.uk/) [24]. The MNI152 standard template space was used as the reference space in the data processing. The image data was processed for brain extraction using the BET (Brain Extraction Tool) [25] and then registered to the 1 mm resolution version of MNI152 template

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(MNI152_T1_1mm) using the FNIRT (FMRIB's Nonlinear Image Registration Tool) [26]. The DTI data were corrected by eddy currents and head motion and then the fractional anisotropy (FA) images were generated by the DTI fitting tool [27]. The FA images were also aligned to the MNI152_T1_1mm as the sMRI.

All sMRI and DTI data downloaded from ADNI dataset were preprocessed by intensity correction with the N3 algorithm [28]. We further process these data following the similar pipeline as that in UKB dataset using the FSL tools and MNI152 template space. Similarly, the BET [25] is used for stripping the non-brain tissues from the source image space, followed by image registration to the MNI152_T1_1mm. The DTI data is preprocessed as below by using PANDA suite 1.2.4 (https://www.nitrc.org/projects/panda/) [29]. Similarly, after correction of eddy-current-induced distortion and simple head-motion, the FA images are extracted using DTI fitting tool [27]. They are aligned and cropped to match with the sMRI data.

To ensure that the inputs of proposed method from different datasets and modalities have the same size, all preprocessed sMRI and FA images are cropped to the size of $160 \times 192 \times 160$ based on a mask generated to remove the uninformative zero-value pixels on the background. The overall data preprocessing pipelines are consistent in the ADNI and UKB datasets, but some details may not be exactly same and subsequent fine-tuning process can solve the minor data gap problem.

III. Proposed Method

This paper proposes a graph transformer geometric deep learning framework to construct and combine the multimodal brain networks from sMRI and DTI data for brain age estimation, as shown in Fig.1. The proposed method consists of a convolutional autoencoder for feature extraction, multi-level construction of brain networks in graph structure, and a graph transformer network to model the multimodal brain networks for brain age estimation, as presented below in details.

A. Convolutional Autoencoder for Feature Extraction

Given the multimodal whole-brain images, a two-stream convolutional autoencoder is proposed to learn their latent representations by the image reconstruction task. Convolution neural network (CNN) consisting of successive convolutional and subsampling layers was proposed to hierarchically learn the features for image recognition [30]. In this work, the convolutional autoencoder (CAE) is a variant of CNN consisting of a convolutional encoder and a symmetrical decoder, as shown in Fig. 2. The encoder aims to learn the latent representation of input images while the decoder reconstructs the input images with representation. Thus, encoder can extract the rich features for reconstruction of input images. The encoder mainly comprises three convolution blocks with each block consisting of two successive $3 \times 3 \times 3$ convolution layers and a $2 \times 2 \times 2$ average pooling layer. Then, a block with two convolution layers is added to further integrate high-level features for image representations. Based on that,

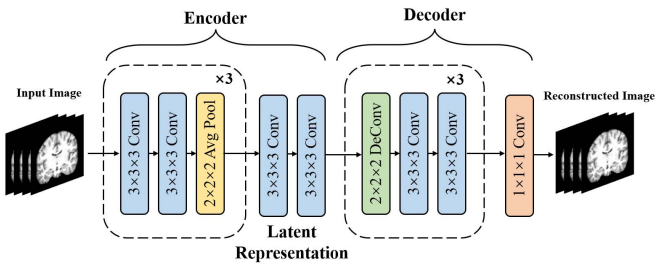

Fig. 2. Architecture of our convolutional autoencoder, which is composed of an encoder to learn the latent representations and a symmetrical decoder for reconstruction of input image.

the decoder uses three deconvolution blocks to reconstruct the input image. Each deconvolution block consists of a $2 \times 2 \times 2$ deconvolution layer followed by two successive $3 \times 3 \times 3$ convolution layers to upsample the feature maps and restore the image details. Finally, a $1 \times 1 \times 1$ convolution layer is added at the end of decoder to predict the image. The channels of the seven blocks are set to 16, 32, 64, 128, 64, 32 and 16, respectively. For the encoder, the channels are gradually increasing, whereas the channels of decoder are gradually decreasing. The convolution layers in the same block have the same number of channels. In addition, the rectified linear unit (ReLU) activation function [31] and the batch normalization (BN) layer [32] are added after each convolution/deconvolution layer except for the last prediction layer, providing non-linear representation and the stability of forward propagation. We set the stride to 1 and apply a zero-padding strategy to make the input and output of convolution layer have the same spatial size. The feature maps to the decoder are used as the image representations for the subsequent brain network construction.

B. Multi-Level Construction of Brain Network

After feature extraction, we propose a multi-level construction of brain networks based on the anatomical regions of interest (ROIs), which are defined by experts in the field of brain science. Thus, the prior knowledge of experts can be embedded to build the topology of brain network and facilitate the interpretation. The brain network is constructed in graph structure with graph nodes and edge connections as shown in Fig. 3. First, to define the brain network nodes, we utilize the Automated Anatomical Labelling (AAL) template [33] with the cerebellum removed, which contains a total of 90 ROIs as the graph nodes, as shown in Fig. 3 (a). The latent representations extracted from the encoder are the feature maps of size $20 \times 24 \times 20 \times 128$. To find the correspondences to the feature maps, the AAL template (on MNI152 space) is cropped to the same size of $160 \times 192 \times 160$ as the input brain image and then downsampled to the same spatial size of $20 \times 24 \times 20$ as that of feature maps. Since three $2 \times 2 \times 2$ average pooling layers are used for downsampling in the autoencoder, the AAL template is also downsampled with three cascaded $2 \times 2 \times 2$ pooling layers. Instead of using the average pooling, we take the label with maximum proportion in the $2 \times 2 \times 2$ window as the output of pooling. Thus, each spatial position of the feature map

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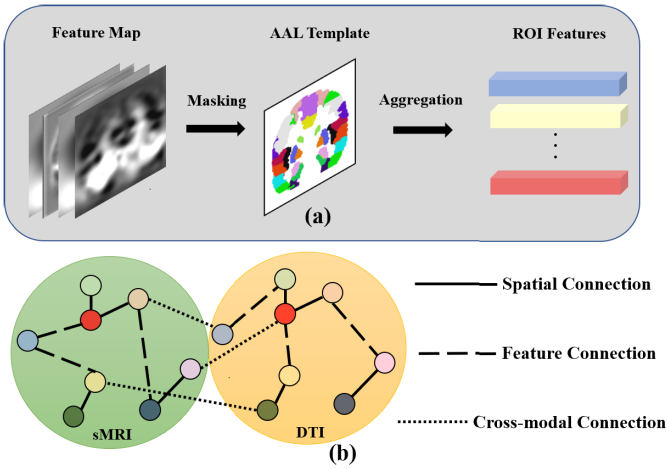

Fig. 3. (a) Generation of node features from ROIs by AAL template masking and feature aggregation; (b) Construction of multimodal brain networks by multi-level connections in terms of spatial relation, feature correlation and cross-modal association.

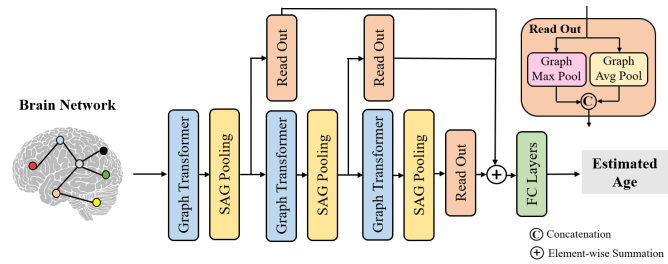

Fig. 4. Architecture of our graph transformer network consisting of three graph transformer layers with each followed by the SAG pooling layer, three readout layers and a fully connected layer, where "Graph Max Pool" and "Graph Avg Pool" denote the aggregations of the selected graph node features by maximizing and averaging, respectively.

has its ROI label. Based on the downsampled AAL template, the feature vectors in the same ROI are averaged to obtain a 128-dimensional feature vector as the node features.

Second, we establish the connections between different nodes, i.e., edges in graph structure, by a multi-level strategy in terms of spatial relation, feature correlation and cross-modal association as shown in Fig. 3 (b). For spatial connection, we compute the ROI centroids of the AAL template by averaging the positions of all points in the ROI, and then calculate the Euclidean distance between two centroids as:

$$D_{ij} = \sqrt{(x_i^c - x_j^c)^2 + (y_i^c - y_j^c)^2 + (z_i^c - z_j^c)^2} \tag{1}$$

where (x_i^c, y_i^c, z_i^c) and (x_j^c, y_j^c, z_j^c) are the centroids of ROI i and j, respectively; $D_{ij} \in \mathbb{R}$ is the Euclidean distance between ROI i and j. According to the spatial relation quantified by Euclidean distances, each ROI node is connected with its K closest ROI nodes. For feature connection, we calculate the cosine similarity between the node features of two ROIs as:

$$S_{ij}^F = \frac{F_i \cdot F_j}{\| F_i \| \| F_j \|} \tag{2}$$

where F_i and $F_j \in \mathbb{R}^{128}$ denote the feature vectors of ROI i and j, respectively; $S_{ij}^F \in \mathbb{R}$ is the feature similarity between ROI i and j. According to the cosine similarities of node features, each ROI node is connected with its K most similar nodes. It is worth noting that the spatial connection is based on the AAL template shared by all subjects while the feature connection is based on the node features of each subject. For each image modality, the spatial and feature connections are combined to build a hybrid connection for each node.

Based on the individual hybrid connections of two modalities, we propose to establish the high-level connections with the cross-modal associations. Given a brain network of one modality, a node connection is a sparse vector with element 1 being connected and 0 being unconnected. We calculate the cosine similarity between the node connections of two modalities as:

$$S_{ij}^C = \frac{C_i^m \cdot C_j^d}{\| C_i^m \| \| C_j^d \|} \tag{3}$$

where C_i^m and $C_j^d \in \mathbb{R}^{90}$ denote the node connections of ROI i in sMRI and ROI j in DTI, respectively; $S_{ij}^C \in \mathbb{R}$ denotes the cosine similarity between the node connections. According to the cosine similarities, each ROI node is connected with the K most similar nodes of another modality. In this work, K is set to 8. Finally, multi-level graph connections are established for the brain networks of two modalities, which facilitates the information aggregation of graph nodes in terms of spatial relation, feature correlation and cross-modal association.

C. Graph Transformer Geometric Learning of Brain Network

Graph neural network (GNN) has achieved great success in representation learning of graph data by iteratively aggregating features of connected nodes [34], [35]. However, most existing GNNs have limitations in learning representation of complex graph with connections of different types and levels. The transformer was designed on fully connected graphs representing all connections between the words in a sequence for natural language processing [36]. Although this architecture can learn the dependencies between connected nodes, it cannot leverage the actual graph connectivity inductive bias [37]. Therefore, we propose a graph transformer network to gradually aggregate the graph representations from the multi-level brain networks for brain age estimation, as shown in Fig. 4.

Motivated by the transformer encoder layer [36], we propose a graph transformer layer with multi-head self-attention mechanism to jointly aggregate the node features for learning the representation of brain networks. The graph transformer layer is composed of multi-head self-attention mechanism and a Feed-Forward Network (FFN) with each followed by a residual connection and layer normalization (LN) [38]. Let $G(O, E)$ represent the multi-level brain network, where $O = \{o_i \in \mathbb{R}^{d_o}\}_{i=1,2,\ldots,n}$ denotes a set of node features, $E = \{e_{ij} = 0 \vee 1\}_{i,j=1,2,\ldots,n}$ denotes the connections between each pair of nodes. In the graph transformer layer l, the node features set O are updated by multi-head self-attention

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mechanism as:

$$Q_i^{h,l} = W_Q^{h,l} o_i^l, \quad K_j^{h,l} = W_K^{h,l} o_j^l, \quad V_j^{h,l} = W_V^{h,l} o_j^l \tag{4}$$

$$w_{ij}^{h,l} = \underset{j \in N_i}{Softmax}\left(\frac{Q_i^{h,l} K_j^{h,l\top}}{\sqrt{d_k}}\right) \tag{5}$$

$$\hat{o}_i^{l+1} = A^l \mathop{\Big\|}_{h=1}^{H} \left(\sum_{j \in N_i} w_{ij}^{h,l} \cdot V_j^{h,l}\right) \tag{6}$$

where $j \in N_i$ denotes the node j connected to the central node i; the weight parameters W_Q, W_K, $W_V \in \mathbb{R}^{d_k \times d_o}$ linearly map node features to different feature spaces Q, K, $V \in \mathbb{R}^{d_k}$ for subsequent correlation modeling and feature aggregation. The weight coefficient $w_{ij} \in \mathbb{R}$ between central node i and its connected node j is calculated by dot product and softmax function. Dot product is a similarity calculation method, and the softmax function can convert the correlations between the connected nodes into weight coefficient. The scale factor $\sqrt{d_k}$ can prevent too large results of dot product from affecting the softmax function. The aggregated central node features can be obtained by weighted summation of connected node features. $h = 1$ to H denotes the attention head number, which means multiple parallel computations to explore different correlations. Large H can capture more diversified correlations, but it increases computation costs. Thus, it is set to 4 in this work. $\|$ denotes the concatenation, the results from different attention heads are concatenated and linearly integrated by the weight parameters $A \in \mathbb{R}^{d_o \times (d_k \cdot H)}$. In this way, the graph transformer layer can aggregate the node features of brain network by multi-head self-attention. The aggregated central node features $\hat{o}_i^{l+1} \in \mathbb{R}^{d_o}$ are then passed to a Feed-Forward Network (FFN), followed by the residual connections and LN layers as:

$$\hat{\hat{o}}_i^{l+1} = LN(o_i^l + \hat{o}_i^{l+1}) \tag{7}$$

$$\hat{\hat{\hat{o}}}_i^{l+1} = W_2^l ReLU(W_1^l \hat{\hat{o}}_i^{l+1}) \tag{8}$$

$$o_i^{l+1} = LN(\hat{\hat{o}}_i^{l+1} + \hat{\hat{\hat{o}}}_i^{l+1}) \tag{9}$$

where $W_1^l \in \mathbb{R}^{d_f \times d_o}$ and $W_2^l \in \mathbb{R}^{d_o \times d_f}$ denote the weight parameters; $\hat{\hat{o}}_i^{l+1}, \hat{\hat{\hat{o}}}_i^{l+1} \in \mathbb{R}^{d_o}$ denote intermediate features; $o_i^{l+1} \in \mathbb{R}^{d_o}$ is the final output features of the central node i. In this work, $d_o = 128$ while d_k and d_f are set to 32 and 64, respectively. The FFN provides non-linear representations for the graph transformer layer, while the residual connections and LN layers ensure the stability of forward information flow and backforward gradient flow.

In a typical CNN model [11], the pooling layer follows the convolutional layer to aggregate the features in local region in a hierarchical downsampling mechanism. Similarly, we apply a Self-Attention Graph (SAG) pooling layer [39] to aggregate node features after each graph transformer layer. SAG pooling considers both node features and graph topology to learn the attention scores of nodes, which determine the nodes to be discarded or kept. The attention scores of graph nodes in the SAG pooling layer l are computed as:

$$Score^l = \sigma(GCL(G^l(O^l, E^l))) \tag{10}$$

where $G^l(O^l, E^l)$ denotes the input brain network of layer l; GCL denotes a graph convolution layer; σ is the $tanh$ activation function. $Score^l$ has the same graph topology as G^l but with only one score value on each node. To better learn the informative features, after sorting the attention scores in decreasing order, the top 80% nodes are kept by the SAG pooling layer for the following layer.

To form a graph representation, we add a readout layer [40] after each SAG pooling layer to aggregate the features of selected graph nodes. The output of readout layer is computed by concatenating the maximum and average of the features over all selected nodes as:

$$R_{out}^l = \underset{i=1}{\overset{n^l}{max}}\, o_i^l \,\|\, \frac{1}{n^l} \sum_{i=1}^{n^l} o_i^l \tag{11}$$

where n^l is the number of nodes in layer l; $o_i^l \in \mathbb{R}^{d_o}$ is the feature vector of i-th node in layer l; $\|$ denotes the concatenation. The readout layer can aggregate the node features to generate a feature vector of fixed size for representation of whole graph brain network. To combine the hierarchical representations of brain network at different levels, we further sum the outputs of three readout layers by residual connection to obtain a graph representation, which is fed into a fully connected layer for brain age estimation.

IV. Experiments and Results

A. Experimental Details

The proposed method is implemented on a single GPU (i.e., NVIDIA GeForce RTX 3090 24 GB), using the PyTorch package in Python. Adam is used as an optimizer with the default learning rate of 0.001. In order to evaluate performance and generalization capability, we conduct the experiments on the preprocessed T1-weighted sMRI and DTI data from UKB and ADNI datasets as described in TABLE II. We first train and evaluate our method on the UKB dataset (16458 healthy subjects), which are randomly divided into training set (14000 subjects), validation set (958 subjects) and test set (1500 subjects). The ages of all subjects (with range of [45, 81]) are split into 7 bins of five-year length. To ensure that the training, validation and test sets have a consistent age distribution over the range, we randomly sample the subjects from each age bin for the three sets based on their proportions in the whole UKB dataset. The trained models with the best performance on the validation set are saved for testing. Then, we transfer the trained model with the UKB dataset to the ADNI dataset (43 healthy subjects). Five-fold cross-validation was performed on the ADNI dataset. In our experiments, three metrics are used for quantitative evaluation of brain age estimation, including mean absolute error (MAE), root means squared error (RMSE) and Pearson correlation coefficient (PCC). The results are presented in terms of mean $\pm$ standard deviation. For UKB dataset, we calculate the mean and standard deviation by testing multiple age estimation models after the training is stable. For ADNI dataset, we calculate the mean and standard deviation of five-fold model results.

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TABLE IV
COMPARISON OF THE BRAIN AGE ESTIMATION RESULTS IN DIFFERENT NODE FEATURES ON UKB DATASET

Method	sMRI			DTI		
	MAE↓	RMSE↓	PCC↑	MAE↓	RMSE↓	PCC↑
Full connection	3.51±0.11	4.69±0.14	0.7993±0.0073	3.78±0.14	4.75±0.15	0.7817±0.0086
Patch-based	3.33±0.16	4.36±0.18	0.8190±0.0098	3.60±0.12	4.58±0.11	0.7941±0.0065
ROI-based	**3.09±0.05**	**3.99±0.10**	**0.8398±0.0067**	**3.45±0.10**	**4.40±0.13**	**0.8004±0.0072**

TABLE V
COMPARISON OF THE BRAIN AGE ESTIMATION RESULTS IN DIFFERENT CONSTRUCTIONS OF BRAIN NETWORK ON UKB DATASET

Method	sMRI			DTI		
	MAE↓	RMSE↓	PCC↑	MAE↓	RMSE↓	PCC↑
CNN + Spatial	3.38±0.11	4.37±0.11	0.8047±0.0078	3.68±0.12	4.71±0.15	0.7833±0.0089
CNN + Feature	3.46±0.17	4.43±0.19	0.7998±0.0094	3.72±0.16	4.80±0.20	0.7791±0.0107
CNN + Hybrid	3.30±0.10	4.21±0.13	0.8210±0.0085	3.57±0.13	4.55±0.14	0.7926±0.0066
CAE + Spatial	3.32±0.09	4.30±0.12	0.8195±0.0069	3.60±0.08	4.61±0.11	0.7885±0.0076
CAE + Feature	3.18±0.12	4.22±0.17	0.8302±0.0090	3.51±0.14	4.48±0.16	0.7972±0.0081
CAE + Hybrid	3.09±0.05	3.99±0.10	0.8398±0.0067	3.45±0.10	4.40±0.13	0.8004±0.0072
Multimodal	MAE↓		RMSE↓		PCC↑	
CNN + Cross-modal	3.01±0.09		4.05±0.14		0.8424±0.0076	
CAE + Cross-modal	**2.71±0.07**		**3.68±0.11**		**0.8682±0.0081**	

Specifically, our proposed method consists of multi-level construction of multimodal brain networks and graph transformer geometric learning of the brain network. In the first part, to learn the latent representations specific to each modality, the two-stream CAEs are individually trained on the sMRI and FA image of DTI, where the mean squared error (MSE) between the predicted and input images is used as the loss function and the minibatch size is set to 4. The feature maps from the encoder are then used for multi-level construction of brain network. In the second part, we train the graph transformer network for brain age estimation. The minibatch size is set to 64 and the MSE between the estimated and chronological ages is used as the loss function. In the following subsections, we first conduct ablation studies to test the effectiveness of these two parts. Then, we compare our method with other methods that are also based on MRI data for brain age estimation. Finally, we explore the relationship between the estimated age difference (EAD) and AD to realize dementia diagnosis.

B. Effectiveness of Multi-Level Brain Network Construction

In this experiment, we test the effectiveness of multi-level brain network construction on brain age estimation with the UKB dataset. The brain network construction depends on the definition of node, calculation of node features, establishment of node connections, and fusion of multimodal brain networks. First, for node definition, the proposed method use the anatomical ROIs based on the AAL template as graph nodes. We compare the ROI-based method to full connection method and patch-based method on the sMRI data. The full connection method uses fully connected layers to concatenate the feature maps of $20\times24\times20$ from the CAE for brain age estimation, denoted as "Full connection". The patch-based method partitions the feature maps into 96 patches of size $5\times4\times5$ as the graph nodes and average pooling is applied to generate the node features, denoted as "Patch-based". The other steps are same as our ROI-based method. Their results are compared in TABLE IV. We can see that the patch-based method achieves better results than full connection method without nodes while the ROI-based method performs better than the patch-based method.

Second, for extraction of node features, the proposed method builds CAE to learn the feature maps of input image, and then generates the feature vector of each ROI based on the AAL template. However, one popular method is to build 3D CNN model to learn features for brain age estimation [11]. For comparison, the CNN model is built with the encoder of CAE followed by a fully connected layer to estimate the brain age, which is trained with the age regression loss. After training the CNN, the output feature map of encoder is used to compute the node features in the same way as CAE. TABLE V compares the results of CAE and CNN-based node features, which are denoted as "CAE" and "CNN", respectively.

Third, we have proposed the multi-level construction strategy to realize the connections of graph nodes by integrating the spatial relation, feature correlation and cross-modal association. Thus, ablation studies are conducted to compare the results of node connections in different levels. In the implementation, the ROI node features are extracted from the above CAE and CNN models while the node connections are established based on the spatial relation, feature correlation, hybrid of spatial and feature connections, and the multi-level connection by cross-modal association, which are denoted as "Spatial", "Feature", "Hybrid", and "Cross-modal" respectively, as shown in TABLE V. It is worth noting that the multi-level connection by cross-modal association is performed to fuse sMRI and DTI data while other connections are performed on single modality. After construction of brain network, the graph transformer geometric learning is performed for brain age estimation.

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TABLE VI
COMPARISON OF THE BRAIN AGE ESTIMATION RESULTS IN DIFFERENT GRAPH LEARNING METHODS ON UKB DATASET

Method	MAE↓	RMSE↓	PCC↑
Baseline	3.87±0.21	4.90±0.25	0.7783±0.0133
GraphSAGE [34]	3.21±0.10	4.13±0.12	0.8275±0.0078
GAT [35]	2.93±0.06	4.01±0.11	0.8528±0.0062
Transformer [36]	3.39±0.14	4.36±0.18	0.8130±0.0095
Transformer + Coding [41]	3.05±0.11	4.08±0.14	0.8419±0.0071
Our Graph Transformer	**2.71±0.07**	**3.68±0.11**	**0.8682±0.0081**

From the results in TABLE V, we can see that the node features from CAE perform better than those from CNN for brain age estimation, especially in brain network by feature connection. The results show the CAE can learn more general-purpose representations of images for brain network construction while the features extracted by CNN are more age-related showing high correlation for different ROIs. As for the node connection, the hybrid of feature and spatial connections performs better than the individual feature and spatial connections in single modality, while the multimodal brain network by cross-modal connection further improves the performance of brain age estimation compared to the single-modal brain networks.

C. Effectiveness of Graph Transformer Network

In this experiment, we test the effectiveness of the graph transformer network on brain age estimation with the UKB dataset. We compare the proposed graph transformer to other graph learning methods including GraphSAGE [34], GAT [35], transformer [36] and Graphormer [41] as well as the baseline by flattening input graphs. Specifically, the baseline method is implemented by a linear regression with the flattened input graphs as features, which cannot make use of the graph structure. GraphSAGE [34] is a general inductive graph learning framework that generates node embeddings by aggregating features from a node's local neighborhood. GAT [35] is a graph attention network with the nodes aggregating their neighborhoods' features with different weights. The standard transformer [36] is implemented based on self-attention mechanism for graph learning. The Graphormer [41] was proposed to encode the structural information of a graph in the standard transformer to better model the graph-structured data. The above methods are implemented based on the released codes with our best efforts. Their results are compared in TABLE VI. We can see that GraphSAGE and GAT performs better than the baseline method by embedding the graph topology in learning the graph features. The Graphormer performs better than the standard transformer by encoding the graph structure. Nevertheless, the proposed method achieves the best performance for brain age estimation because of its better modeling the multimodal multi-level brain graph networks.

In our graph transformer network, the number of self-attention heads and the ratio of discarded nodes in the SAG pooling are two important parameters for brain age estimation. Thus, experiments are conducted to test the brain age estimation performances by varying the head number

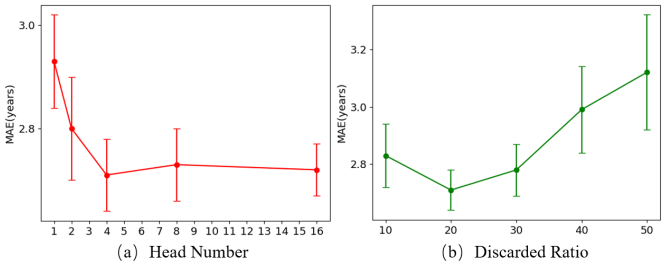

Fig. 5. Results of variant hyperparameters: (a) head number in graph transformer layer and (b) ratio of discarded nodes in SAG pooling layer.

TABLE VII
COMPARISON OF THE BRAIN AGE ESTIMATION RESULTS IN DIFFERENT MULTIMODAL FUSION STRATEGIES ON UKB DATASET

Method	MAE↓	RMSE↓	PCC↑
Early Fusion	3.42±0.11	4.44±0.12	0.8159±0.0077
Feature Fusion	3.28±0.05	4.18±0.04	0.8244±0.0056
Hybrid Fusion [42]	3.06±0.12	4.01±0.15	0.8435±0.0086
Late Fusion	3.12±0.09	4.16±0.13	0.8364±0.0074
Proposed Fusion	**2.71±0.07**	**3.68±0.11**	**0.8682±0.0081**

from 1 to 16 and varying the discarded ratio from 10% to 50%. The MAE results are shown in Fig.5. Fig.5 (a) shows that MAE is reduced by increasing the head number from 1 to 4, but further increasing cannot reduce MAE but increase the network complexity and computation costs. Fig.5 (b) shows that the MAE is reduced by increasing the ratio of discarded nodes to 20% but it is increased by further increasing. Discarding a small proportion of graph nodes can reduce the noise and redundancy of graph to make the model better learn informative features while discarding too many nodes will result in losing important information and reducing the performance. Thus, the head number is set to 4 and the ratio of discarded nodes is set to 20% in our experiments.

Finally, as for the fusion of two modalities, the proposed method constructs multimodal brain network, followed by graph transformer geometric learning to aggregate multimodal representations for brain age estimation. We compare the proposed fusion strategy to the early fusion, feature fusion, late fusion and hybrid fusion in [42], which are denoted as "Proposed Fusion", "Early Fusion", "Feature Fusion", "Late Fusion" and "Hybrid Fusion", respectively, as shown in TABLE VII. In our experiments, the early fusion is implemented by directly concatenating the sMRI and FA images as inputs, followed by a 3D CNN for brain age estimation. We implement the feature fusion by taking out the features after global average pooling of two separate 3D CNNs trained on sMRI and FA images by a fully connected layer and fit a linear regression. The hybrid fusion is implemented by combining the element-wise summation, element-wise product and element-wise maximization for fusion of the multimodal features from the 3D CNNs as in [42]. The late fusion is implemented by averaging the age estimation results from our method trained on sMRI and DTI separately. The results in TABLE VII show that the hybrid fusion performs better than

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TABLE VIII
RESULTS COMPARISON OF OUR PROPOSED METHOD WITH OTHER METHODS IN BRAIN AGE ESTIMATION

Method	Modality	MAE↓	RMSE↓	PCC↑
Feng et al., 2020 [8]	sMRI	3.54±0.16	4.60±0.19	0.7923±0.0107
Jiang et al., 2020 [9]	sMRI	3.65±0.12	4.71±0.13	0.7834±0.0082
Bashyam et al., 2020 [10]	sMRI	3.44±0.13	4.52±0.16	0.8021±0.0079
Peng et al., 2021 [11]	sMRI	3.29±0.12	4.30±0.10	0.8215±0.0060
Cheng et al., 2021 [12]	sMRI	3.10±0.07	4.03±0.09	0.8376±0.0075
He et al., 2021 [15]	sMRI	3.16±0.10	4.09±0.14	0.8342±0.0084
Proposed Method	**sMRI**	**3.09±0.05**	**3.99±0.10**	**0.8389±0.0067**
Proposed Method	**sMRI+DTI**	**2.71±0.07**	**3.68±0.11**	**0.8682±0.0081**

the early, feature and late fusions while the proposed fusion achieves the best performances.

D. Comparison With Other Methods

TABLE VIII compares the proposed method with other methods recently published in the literature [8], [9], [10], [11], [12], [15]. The compared methods are based on deep neural networks including the 2D CNN [10], 3D CNN [8], [9], [11], [12] and transformer [15] for brain age estimation using T1-weighted sMRI data. For fair comparison, all methods are performed with the same training configuration on the same training, validation and testing datasets. In the experiments, we implement the methods [10], [11], [12], [15] with their released source codes on the github. As for the methods without released codes [8], [9], since the 3D CNN models are built on the typical architecture and easy to reproduce, we implement these methods from the details of their papers with our best efforts.

The results in TABLE VIII show that the 3D CNN method [11] performs better than the 2D CNN method [10] and the typical 3D CNN architectures [8] and [9]. The two-stage cascade 3D CNN and global-local transformer can further improve the results of brain age estimation compared to the 3D CNN method [11]. Nevertheless, our proposed method performs better than all other methods using sMRI data. By fusing sMRI and DTI data, our proposed method can further improve the performances by reducing 0.38 of MAE. It is worthy to note that the results in Table VIII may be different from those reported in the literature because of the different studied data and training configuration.

E. Estimated Age Difference and Dementia Diagnosis

The final experiment is to transfer our deep network model pre-trained on the UKB dataset to the ADNI dataset, and explore the association between EAD and dementia diagnosis. To train or fine-tune the proposed model, five-fold cross-validation is performed on the NC subjects of ADNI dataset. TABLE IX compares the results of the proposed model without pre-training, the pre-trained model without fine-tuning, and fine-tuning the pre-trained model on the NC subjects, denoted as "NC#", "NC*", and "NC", respectively. We can see that fine-tuning the pre-trained model performs better than other two models, showing the effectiveness of transfer learning. Furthermore, we fine-tune the pre-trained model with all NC subjects and test it on the MCI and AD subjects of ADNI dataset. Their results of MAE, RMSE, PCC and EAD are shown in TABLE IX. The EAD increases as subjects progress from NC, MCI to AD, indicating that it is an important biomarker for evaluation of dementia severity and can help early diagnosis of AD.

TABLE IX
THE BRAIN AGE ESTIMATION RESULTS ON AD, MCI AND NC SUBJECTS OF ADNI DATASET WHERE "NC#", "NC*" AND "NC" DENOTE THE PROPOSED MODEL WITHOUT PRE-TRAINING, THE PRE-TRAINED MODEL WITHOUT FINE-TUNING, AND FINE-TUNING THE PRE-TRAINED MODEL ON THE NC SUBJECTS, RESPECTIVELY

Group	MAE↓	RMSE↓	PCC↑	EAD
NC#	3.91±0.14	5.04±0.19	0.7479±0.0086	0.61±4.46
NC*	3.67±0.30	4.71±0.35	0.7763±0.0134	-0.32±4.02
NC	2.63±0.15	3.59±0.14	0.8458±0.0063	-0.02±3.29
MCI	5.59±1.62	7.48±1.57	0.3263±0.0452	2.99±6.86
AD	7.77±1.35	9.50±1.42	0.1552±0.0395	5.12±6.17

TABLE X
THE CLASSIFICATION RESULTS OF AD VS. NC, MCI VS. NC AND AD VS. MCI ON ADNI DATASET

Classification (%)	ACC↑	SEN↑	SPE↑
AD vs. NC	85.9±2.8	79.8±6.7	92.8±6.6
MCI vs. NC	74.5±2.0	71.2±4.3	78.9±5.8
AD vs. MCI	65.8±3.1	59.3±5.9	71.2±4.3

In addition, we concatenate the EAD and the learned feature vector from the graph transformer network as the inputs of multi-layer perceptron (MLP) for AD diagnosis. The group with severer dementia is considered positive, while the other group is negative. The model is trained by cross-entropy loss and Adam optimizer with the learning rate of 0.001 and the minibatch of 4. Five-fold cross-validation is adopted and the accuracy (ACC), sensitivity (SEN) and specificity (SPE) are computed for evaluation. The classification results of AD vs. NC, MCI vs. NC, and AD vs. MCI are shown in TABLE X. The proposed method achieves ACCs of 85.9% and 74.5% on average for AD and MCI diagnosis, respectively. Different from the existing classification methods, the proposed method only utilizes the NC subjects for training the model to learn the EAD and age-related features for disease diagnosis.

V. DISCUSSION

We have proposed a graph transformer geometric learning framework for brain age estimation with sMRI and DTI data. Recently, the deep learning methods have been widely investigated on computational analysis of MRI data for brain age estimation [8], [9], [10], [11], [12], [15], which need less preprocessing steps and performs better than the traditional machine learning based methods [6], [7]. Compared to these deep learning studies, our method can efficiently model the cross-modal interaction of sMRI and DTI data and fuse their multi-level features in a framework to improve brain age estimation, and has the following differences. First, instead of building a 3D CNN model on the whole brain image to learn a global representation for brain age estimation [8], [9], [11], [12], two-stream CAE is built to learn the latent representations of each modality, followed by construction of

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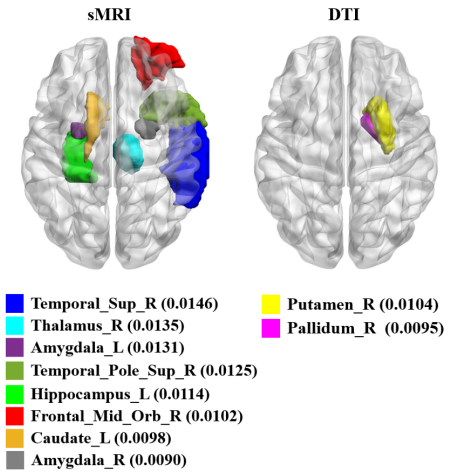

Fig. 6. Visualization of top 10 ROI nodes in the learned brain network with their weights calculated by attention scores.

a multi-level graph network using the anatomical ROIs as nodes and a hybrid connection strategy between nodes in intra-modality and inter-modality. This graph network can not only learn the representation of local regions but also model the global topology structure of whole brain to provide more powerful representation of brain images. Second, different from the global-local transformer method which applied the self-attention mechanism to combine the global and local features learned by 2D CNN from whole brain and local patches [15], we extract the local features from the anatomical ROIs and construct the multi-level graph network to model the global topology structure of ROIs, followed by the graph transformer geometric learning for hierarchical feature aggregation and brain age estimation. Thus, the proposed method can not only make full use of global and local features for more accurate estimation of brain age but also facilitate the interpretation with ROIs. Finally, based on the proposed model, we have explored the association between the EAD and dementia for early AD diagnosis. The results in TABLE IX shows that the EAD is a promising biomarker for dementia classification.

In addition to brain age estimation, interpretation is also important for brain image analysis. To facilitate interpretation, we output the attention scores of ROI nodes from the first SAG pooling layer of the graph transformer network. The softmax function is used for normalization of the scores for each subject, then we average the attention scores over all subjects in the UKB testing set. The average scores of all ROI nodes are sorted in descending order and the top 10 ROI nodes are selected for visualization by BrainNet Viewer [43], as shown in Fig. 6. We can see that there are 8 ROIs selected from sMRI and 2 ROIs from DTI, showing that sMRI is more important than DTI for brain age estimation. Most ROIs are associated with aging-induced brain atrophy, and the selected ROIs such as Hippocampus and Amygdala are also involved in diagnosis of dementias such as AD.

Our deep learning model is trained with sMRI and DTI data from 14000 subjects of UKB dataset, which takes 32 hours to achieve the convergence using a single NVIDIA GeForce RTX 3090 GPU. The inference time is fast in about 0.015 second per subject. It takes 1 hour for fine-tuning the pre-trained model on the ADNI dataset.

Although our method has achieved good performance in brain age estimation and AD diagnosis, there are several limitations to be addressed in the future. First, in the proposed method, the construction of brain network and graph transformer network are trained separately which may lead to sub-optimal performance. An end-to-end deep learning framework can be investigated to improve performance. Second, the proposed method cannot deal with the missing modalities for multimodal brain image analysis, limiting the datasets used for study. In the future, the imputation technologies can be used to improve this method.

VI. Conclusion

This paper proposes a graph transformer geometric learning framework for brain age estimation using multimodal brain images, which can model the cross-modal interaction and hierarchical fusion of multimodal features. The proposed method has been tested on the sMRI and DTI data from 16458 UKB subjects. We also transfer the pre-trained model to the ADNI dataset for brain age estimation and dementia diagnosis. The experimental results and comparison demonstrate the promising performance of our proposed method.

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